

STT LATENCY BENCHMARK ACROSS SIX ENGINES

We benchmarked six STT options available through Telnyx using the same 3.70 second English audio sample. Each model ran 10 times with realtime audio streaming and a 1 second prewarm. The main number is EOU latency: time from speech end to the final transcript a cascaded voice AI system can act on.

FASTEST EOU P50

396 ms

Nova-3 had the lowest EOU p50 in this run. Flux was close at 416 ms. Speechmatics also landed under 500 ms p50.

Deepgram

Nova-3

396 ms

EOU p50

Deepgram

Flux

416 ms

EOU p50

Speechmatics

Speechmatics

455 ms

EOU p50

AssemblyAI

AssemblyAI

556 ms

EOU p50

xAI

Grok STT

560 ms

EOU p50

Soniox

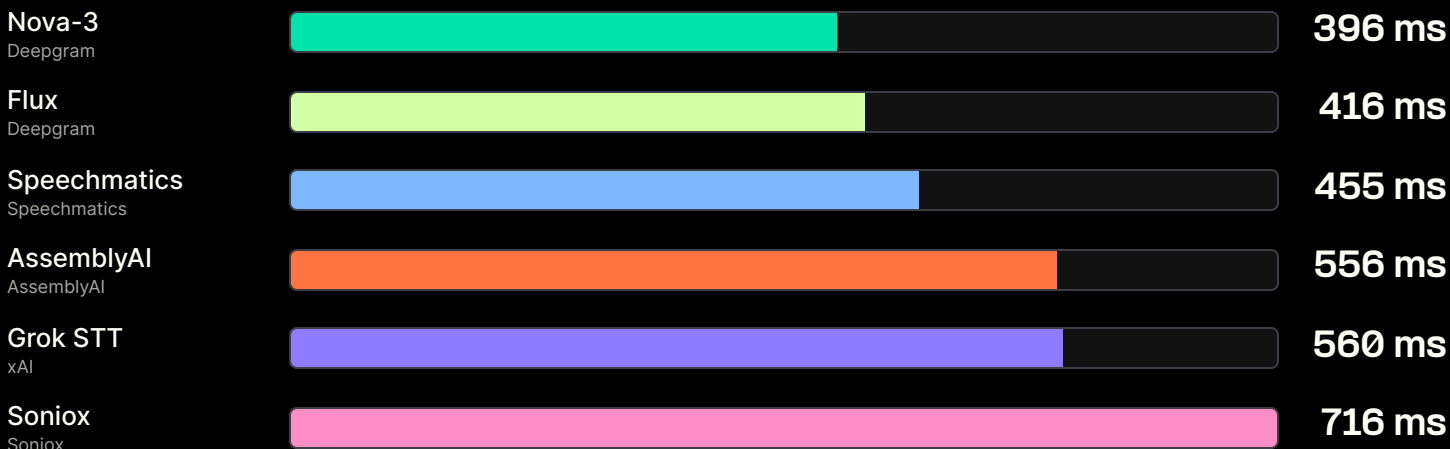
Soniox

716 ms

EOU p50

EOU P50 COMPARISON

All six engines returned complete usable turns in 10/10 runs. The fastest EOU p50s were Nova-3, Flux, and Speechmatics.



MAIN METRIC

EOU p50

Median time from speech end to the final transcript containing the complete expected turn.

FINAL TIMING

EOU p95

Tail version of the same metric. It shows how slow the slower runs got.

STREAMING RESPONSE

First interim

Time until the first non-final transcript appears while the user is still speaking.

SERVICE ESTIMATE

Service p50

EOU p50 after subtracting one measured WebSocket round trip.

READOUT

All finals landed

Every engine returned complete usable turns in 10 out of 10 runs.

FASTEST GROUP

Under 500 ms

Nova-3, Flux, and Speechmatics all finished below 500 ms EOU p50.

MIDDLE GROUP

Near 560 ms

AssemblyAI and Grok STT were close on EOU p50 in this run.

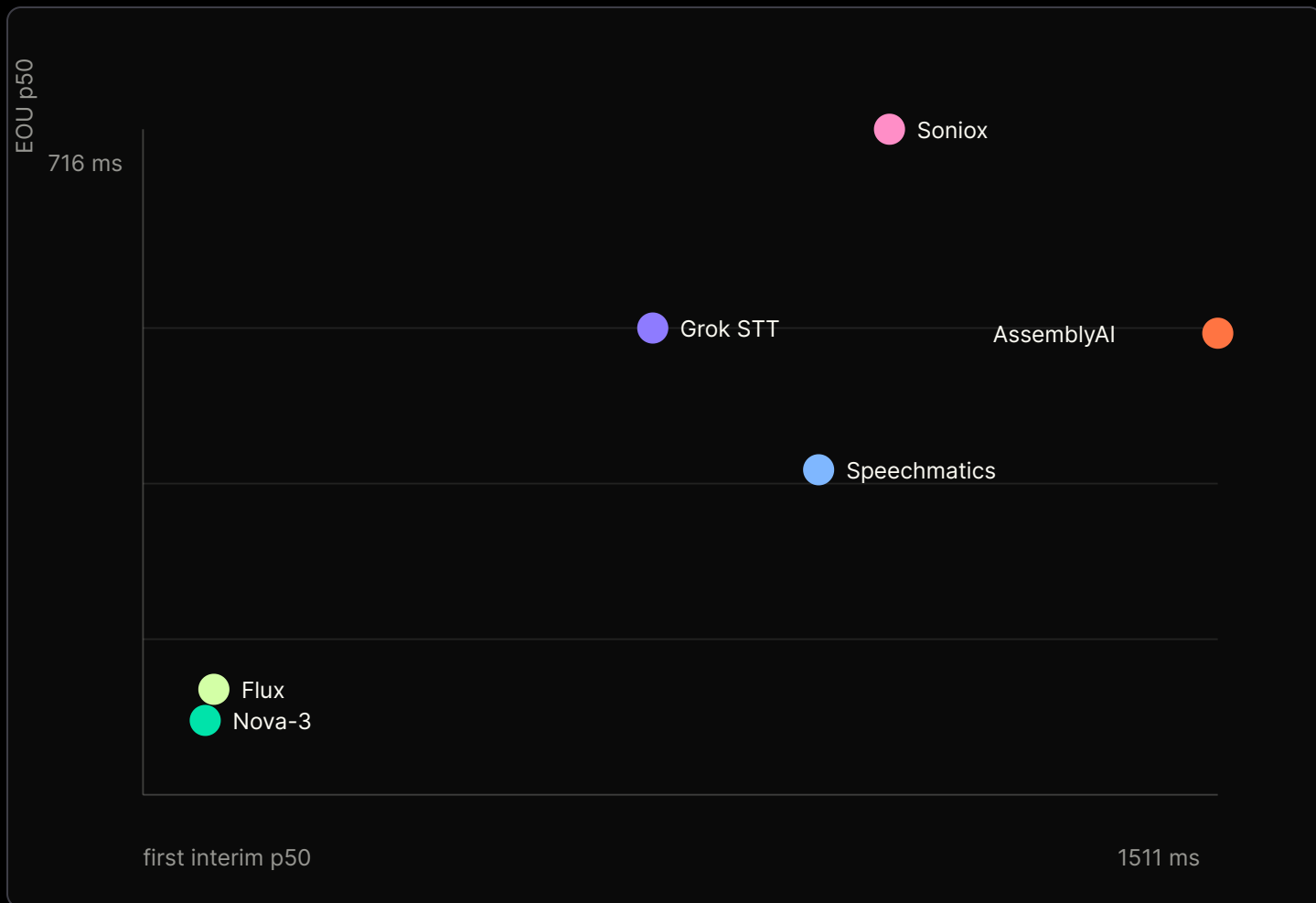
TAIL NOTE

Soniox outlier

Soniox had one high-tail run, which is why the p95 needs care.

FIRST INTERIM AND EOU TIMING

The engines cluster closer on EOU latency than they do on first interim. That matters when deciding between agent handoff and visible transcription use cases.



BEST REGION

Lower left

Lower EOU and lower first interim means the provider is faster on both axes.

VOICE AGENT HANDOFF

Y axis matters

EOU is the more important latency for deciding when the agent can respond.

LIVE CAPTIONS

X axis matters

First interim controls how quickly visible partial text starts appearing.

INTERPRETATION

Different jobs

A provider can be fine for handoff even if partial text appears later.

WHAT THE RUN SAYS

Speechmatics was competitive on EOU. AssemblyAI and Grok STT were close to each other. Soniox worked after endpointing configuration, but had the slowest p50 and one high-tail outlier.

FASTEST EOU

Nova—3 and Flux stay close

Nova-3 finished at 396 ms EOU p50. Flux finished at 416 ms.

NEW PROVIDER RESULT

Speechmatics is competitive

Speechmatics finished at 455 ms EOU p50 and 476 ms EOU p95.

CONFIG-SENSITIVE RESULT

Soniox needs endpointing

Soniox returned usable finals with endpointing=500 and trailing silence, but p95 was 2070 ms.

MODEL	ENGINE	EOU P50	SERVICE P50	FIRST INTERIM P50	EOU P95	TURNS
● Nova-3	Deepgram	396 ms	290 ms	88 ms	419 ms	10/10
● Flux	Deepgram	416 ms	302 ms	98 ms	450 ms	10/10
● Speechmatics	Speechmatics	455 ms	358 ms	970 ms	476 ms	10/10
● AssemblyAI	AssemblyAI	556 ms	453 ms	1511 ms	608 ms	10/10
● Grok STT	xAI	560 ms	442 ms	721 ms	674 ms	10/10
● Soniox	Soniox	716 ms	620 ms	1067 ms	2070 ms	10/10

TEST SETUP

Same fixture

3.70 second English sample, realtime audio streaming, 10 runs per model.

WARM STATE

1s prewarm

The timer starts after prewarm, so the prewarm silence is not counted.

CONFIG NOTE

AssemblyAI en—US

AssemblyAI used language=en-US for this English fixture.

CONFIG NOTE

Soniox endpointing

Soniox used endpointing=500 plus trailing silence to return complete finals.

WHAT THIS MEANS FOR USE CASES

Use the benchmark as a decision aid, not a universal ranking. The right metric depends on what the product needs the transcript to do.

CASCADED VOICE AI

Watch EOU p50

This is the handoff point from STT to the next system. Nova-3, Flux, and Speechmatics led this run.

VISIBLE LIVE TEXT

Watch first interim

Nova-3 and Flux showed the fastest first interim p50. AssemblyAI, Soniox, and Speechmatics were later.

PROVIDER TESTING

Check config first

AssemblyAI and Soniox both changed materially when run with the right engine-specific settings.

USE CASE	METRIC	BEST RESULT HERE	READOUT
Cascaded voice AI handoff	EOU p50	Nova-3, 396 ms	Flux and Speechmatics are also under 500 ms p50.
Visible live transcription	First interim p50	Nova-3, 88 ms	Flux is close at 98 ms. Other providers start visible partials later.
Tail behavior	EOU p95	Nova-3, 419 ms	Soniox had one outlier in this run, so p95 needs more sampling before a sales claim.
Provider comparison	Config parity	Engine-specific defaults	Run provider-specific settings before calling a model slow or broken.

TEST SETUP

Same fixture

3.70 second English sample, realtime audio streaming, 10 runs per model.

WARM STATE

1s prewarm

The timer starts after prewarm, so the prewarm silence is not counted in latency.

METRIC SOURCE

Runner output

Numbers use the aggregate EOU p50 and EOU p95 from the benchmark run.

SALES CAVEAT

Do not overclaim

Use this sample as directional data. Customer audio should be tested before final provider choice.